iMAppGAN: Integrated Motion Appearance Generative Adversarial Networks for Video Anomaly Detection

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Abstract. The identification of anomalies in videos primarily focuses on detecting rare or inappropriate occurrences within specific contexts. The other state-of-the-art models often rely on either future frame prediction or reconstruction approaches. However, these methods exhibit limitations, such as inadequate detection due to the less variation between the generated normal and anomaly video frames. Thus, there is a need that combine the strengths of both future frame prediction and reconstruction models, the Integrated Motion-Appearance Generative Adversarial Networks (iMAppGAN). It is an end-to-end network that sequentially performs the prediction of future frames and further reconstructs the predicted video frame. The prediction of future frames facilitates the detection of abnormal events by emphasizing significant reconstruction errors. This is achieved through the integration of an autoencoder block and a UNET block within the generator to generate the normal video frame and distort the anomaly video frame. The first autoencoder block features a dual-stream encoder designed for extracting both motion and appearance features while the UNET block is responsible for reconstructing the predicted future frames received from the autoencoder. This results in a robust solution for effectively detecting anomalies in video surveillance, overcoming the drawbacks of both the prediction and reconstruction methods, and enhancing overall performance. Experimental results show that our novel iMAppGAN outperforms existing state-of-the-art models, demonstrating remarkable performance with an AUC score of 97.9% on UCSD Ped2, 90.8% on CUHK Avenue, and 75.3% on the ShanghaiTech dataset.

Keywords: Generative adversarial networks (GAN) · Adversarial learning · One-class classification (OCC) · Video anomaly detection

1 Introduction

The surge in video monitoring data, driven by widespread video surveillance system adoption, poses challenges in the continuous and error-prone task of

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human monitoring. Anomaly Detection (AD) plays a pivotal role, particularly in video surveillance and industrial inspection applications. The manual monitoring of video feeds proves labor-intensive and error-prone resulting in increasing demand for intelligent surveillance systems. These systems autonomously identify anomalies, mitigating the limitations of human monitoring. The definition of anomalies is context-dependent, often signifying infrequent occurrences deviating from regular events. A significant challenge for AD emerges due to the diversity and rarity of anomalous instances in real-world datasets.

Consequently, AD is commonly performed within a one-class learning framework, exclusively training on normal data. This approach aims to achieve increased discrimination between normal and anomaly samples, even with the meager number of abnormal samples. The latest advancements in deep learning have significantly enhanced unsupervised approaches by introducing reconstruction-based methods.

A reconstruction-based AD consists of a network trained exclusively on normal samples that struggles to extract anomalous data efficiently, relying on the estimated reconstruction error for detection. Autoencoder (AE) methods encode the input, extract significant features, and decode to reconstruct inputs, using the encoding as a bottleneck for capturing high-dimensional patterns. Hasan et al. [9] utilizes AE for image reconstruction, identifying anomalies by comparing each pixel of the reconstructed image to that of the input image.

Taking temporal features into account [24], prediction-based approaches introduce the concept of generating missing frames, such as middle frames [34] or future frames [15], based on motion continuity. The dissimilarity between the predicted and its corresponding real video frame is then used to infer the likelihood of the occurrence of an anomaly. To generate the future predicted frame from a window of preceding video frames, predictive models have grabbed a lot of attention, particularly in applications like autonomous driving and video understanding. For example, Liu et al. [15] introduce a method employing the U-Net architecture to forecast future frames. The predicted future video frame is further used to find anomalies in the video [25]. In the past few years, Generative Adversarial Networks (GANs) have gathered substantial attention for their remarkable capabilities in generating realistic data distributions.

Establishing a suitable threshold for reconstruction errors is challenging when determining abnormal events in a video frame. This challenge stems from the unreliability of reconstruction algorithms to consistently yield significant errors for anomalies. Prediction methods can set normal and abnormal video frames apart, but their sensitivity to motion-related noise, such as optical flow, makes them less robust than reconstruction methods. Thus, there is a need for a framework that takes advantage of both reconstruction and prediction methods.

We present a novel approach to combine future video frame prediction and reconstruction-based methods that utilize the Adversarial training of GAN. Our prediction model is comprised of Wideresnet [32] which takes the sliding window of video frames as input and segregates it into two streams to handle the motion and appearance by utilizing the temporal shift module [12]. Inputting a sequence of frames $(T_1, T_2, ..T_n)$ into the prediction network, it extracts

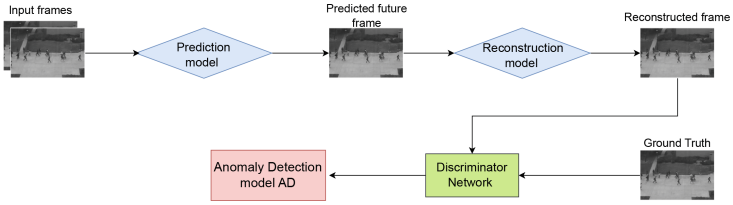

Fig. 1. Training Pipeline for iMAppGAN. A sliding window of video frames is provided as input to the prediction model which predicts the future video frame in the sequence and passes it on to the reconstruction model where the predicted future frame is reconstructed and then fed into the discriminator to identify it as real or fake. Here, ground truth is the next frame in the input sequence.

features from preceding input video frames for predicting the subsequent frame. The outcome of the prediction network is further passed to the UNET-based reconstruction network. Thus, we utilized an autoencoder which will reconstruct the predicted future frame and pass it to the discriminator which then discriminates between the original video frame and the reconstructed future video frame. Figure 1 provides an overview of our methodology, iMAppGAN.

Our key contributions are outlined below:

1. The video AD problem is effectively handled by the framework that integrates the prediction and reconstruction methods. This technique improves the capability to differentiate between normal and anomalous occurrences by overcoming the shortcomings due to reconstruction errors.
2. The prediction network in our GAN-based model incorporates a two-stream approach to extract motion and appearance information whereas, the reconstruction network further enhances the future frame prediction of normal events.
3. Our framework incorporates adversarial learning to handle complex scenes, preserving discriminative capabilities and enhancing generalization for an expanding range of normal events.

The remaining sections of the paper are structured in the following manner: An overview of earlier work in video AD is given in Sect. 2. Further, Sect. 3 elaborates on the proposed iMAppGAN framework, and Sect. 4 presents a practical analysis of the model. Finally, the paper concludes in Sect. 5, addressing the future work.

2 Related Work

The video AD faces various challenges that have been identified over the years. Traditional methods such as modeling graph representations [1], probabilistic models like Markovian Random Field (MRF) [11], and normal trajectory tracking [31] have been used as AD. The aforementioned methods usually fail to adapt

to the different anomaly characteristics and are unable to differentiate between complex patterns leading to the utilization of manual feature engineering. Video AD has been significantly improved with the help of deep learning (DL) methods [25], as they can learn dependencies and complex patterns more reliably and efficiently without the need for external supervision. Two cutting-edge DL-based methods for video-AD are the reconstruction method and the prediction method, which learn from the normal events.

2.1 Reconstruction Based Methods

Anomalies cannot be reconstructed by using just the normal video frames. Hence, the reconstruction-based methods utilize the distribution of video frames to learn it and identify the video frame having high reconstruction errors as anomalies. Several works show that AEs are utilized for AD, such as convolutional AE [9]. The input sequence of video frames can be reconstructed from training video frames by Conv-AE (Hasan et al. [9]), which utilizes a deep AE. The encoding serves as a data bottleneck, aiding the network in capturing the high-dimensional features from the normal video frames. Although reconstruction-based methods have been used for video AD, their lack of generalization often makes it difficult to discriminate abnormal from normal samples. This difficulty arises from the inherent tradeoff between achieving accurate reconstruction and maintaining effective discrimination capabilities.

2.2 Prediction Based Methods

Predictive models aim to generate future video frames based on the features from the sequence of preceding frames. For instance, in the context of detecting video forgery, a recurrent autoencoder uses Long Short-Term Memory (LSTM) networks [4] to capture temporal correlations among patches from a series of input video frames. Building on the advancements in video prediction techniques [21], some research integrates predictive concepts into anomaly detection [15]. The architecture of ASTNet [12] is structured as an autoencoder, with the primary objective of efficiently incorporating both temporal and spatial features to predict the future video frame.

2.3 Generative Adversarial Networks

Deep learning has experienced a significant breakthrough with the development of Generative Adversarial Networks (GAN) [8]. GAN operates on the principle that the generator aims to produce authentic data while the discriminator strives to distinguish real frames from the generated video frames. Lee et al. [13] used a dual-directional generator with Long Short-Term Memory (LSTM), capturing both temporal and spatial properties of normal video frames. Song et al. [28] proposed Ada-Net for capturing the long-term dependencies from the video by integrating both AE and GAN. Further in Sect. 3, we discuss the proposed method iMAppGAN which integrates the prediction method and the reconstruction method with adversarial learning for video AD.

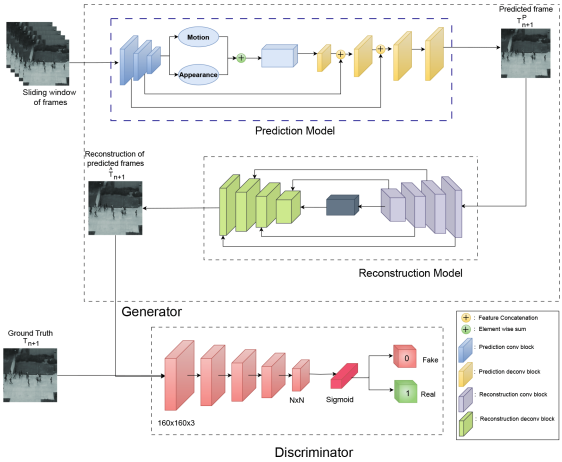

Fig. 2. Overall pipeline of iMAppGAN framework.

3 Methodology

The integrated Motion Appearance GAN (iMAppGAN) model comprises two main networks namely, the Generator and the Discriminator. It is essentially an integration of a prediction-based adversarial AE model and a reconstruction method utilizing adversarial learning. The complete framework is depicted in Fig. 2.

3.1 Generator

The Generator $\mathcal{G}$ network consists of two models, first, we have a Prediction Model $\mathcal{G}1$ and then a Reconstruction Model $\mathcal{G}2$. The prediction model and the reconstruction model are based on autoencoder (AE) and UNET, respectively, as explained below:

Prediction Model. The first block of our Generator, i.e., $\mathcal{G}1$, consists of an encoder that extracts features related to both motion and appearance from the input video frames using a two-stream channel. The encoder employs *WiderResnet* [32] as its backbone. Videos come with both motion and appearance features leading us to split the final layer of our deep CNN into two streams. Further, the *temporal shift module* is utilized to efficiently capture information related to motion from the video frames while for the appearance branch, we concatenate the inputs to maintain spatial consistency. Finally, all the features are concatenated and passed to the decoder. The decoder upsamples the low-dimensional latent space back to input resolution by the process of deconvolution. The framework involves a 1×1 convolution layer to minimize the number of parameters accompanied by a sequence of DeConv blocks, as shown in Fig. 3. It is then

succeeded by a Channel-Attention (CA) module, which improves our model by exploiting the channel relationships between features. The training of the $\mathcal{G}1$ is further stabilized by concatenating the generated feature maps from this layer with encoder features using skip connections.

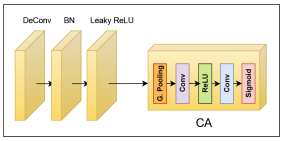

Fig. 3. The decoder in the prediction model employs attention-based deconvolution to predict future frames. The Deconvolution block consists of Leaky-ReLU activation layers, Batch Normalization (BN), DeConv, and a channel attention (CA) module aimed at enhancing the cognitive capabilities of our model.

Reconstruction Model. The subsequent UNET block, $\mathcal{G}2$, in the sequence, receives the up-sampled images that were obtained from $\mathcal{G}1$. The convolution block in UNET receives the predicted future frame as input. The significant features of the input video frame are retrieved and then compressed into a compact section known as the bottleneck before passing to the deconvolution block. The bottleneck output is sent to the deconvolution block, which is made up of several up-sampling modules to generate the video frame. Further, the skip connections are introduced between the convolution and deconvolution blocks to avoid the issue of information imbalance and gradient vanishing.

3.2 Discriminator

The discriminator $\mathcal{D}$ model distinguishes between the generated video frame and the corresponding real video frame. It learns to differentiate the ground truth (T_{N+1}) from the reconstructed predicted video frame ($\hat{T}_{N+1}$). The reconstruction error bridges the gap between real and fake which signifies the regular and abnormal frames. Since the generator is only trained from normal samples, it results in low reconstruction loss of the normal video frame as compared to the anomaly video frame. As a result, the discriminator learns to differentiate between normal and anomalous video frames. The discriminator uses a convolutional architecture model called PatchGAN, proposed by Isola et al. [10]. It compares the reconstructed future frame against the actual video frame at the patch level. The result is a matrix divided into $N \times N$ patches, each corresponding to a particular pixel of the video frame.

3.3 Adversarial Training

The adversarial training is utilized to train both the generator and the discriminator models using a min-max game. The discriminator network differentiates

between the real and generated video frames. The training pipeline is shown in Fig. 1.

Discriminator Training. The binary classification between the fake and real video frames is the aim of training the discriminator. The generated (false) video frame $\hat{F}_{t+1}$ is categorized as 0, while the ground truth F_{t+1} is classified as 1. The loss function for the true and false classifications is shown in Eq. 1.

$$L_{adv}^{\mathcal{D}}(\hat{T}, T) = \sum_{x,y} L_{BCE}\left(\mathcal{D}(T)_{x,y}, 1\right) + \sum_{x,y} L_{BCE}\left(\mathcal{D}(\hat{T})_{x,y}, 0\right) \tag{1}$$

Here x and y denote the image's 2D-pixel locations, and L_{BCE} denotes the Binary Cross Entropy loss which is defined in Eq. 2:

$$L_{BCE}(\hat{Z}, Z) = -\frac{1}{n}\sum_{j=1}^{n} Z_j \cdot \log \hat{Z}_j + (1 - Z_j) \cdot \log\left(1 - \hat{Z}_j\right) \tag{2}$$

Here $\hat{Z} \in [0, 1]$ and Z is either 0 or 1.

Generator Training. The generator network's training aims to deceive the discriminator network into recognizing the generated video frame as real. For training $\mathcal{G}$, the adversarial loss is represented by Eq. 3:

$$L_{adv}^{\mathcal{G}}(\hat{T}) = \sum_{x,y} L_{BCE}\left(\mathcal{D}(\hat{T})_{x,y}, 1\right) \tag{3}$$

When the samples contain values far apart from the input space Y, $\mathcal{G}$ can potentially generate samples capable of deceiving $\mathcal{D}$. Similar to [17], we tackle this issue by incorporating the structural-similarity loss. A combined loss of the adversarial loss (Eq. 3) and the L_2 loss (Eq. 4) to train $\mathcal{G}$.

$$L_{int}(\hat{T}, T) = \|\hat{T} - T\|_2^2 \tag{4}$$

Therefore, we need to minimize $\lambda_{\text{int}}\, L_{int}\left(\hat{T}_{r+1}, T_{r+1}\right) + \lambda_{adv} L_{adv}^{\mathcal{G}}\left(\hat{T}_{r+1}\right)$ Further, there is a need to maintain a balance between image similarity and quality in comparison to ground-truth video frame samples [21].

Image Gradient Constraints. To enhance the quality of the predicted video frame as well as the reconstructed video frame, Mathieu et al. [21] exploited gradient and intensity differences. Our proposed method aims to predict, given an input sliding window of video frames $(T_1, T_2, ..., T_j)$, the predicted future video frame T_{j+1}^{p} and then reconstruct it to generate $\hat{T}_{j+1}$. A gradient constraint is introduced to the video frame to address the potential blur when utilizing L_2 distance. Below Eq. 5 shows the image gradient loss.

$$L_{gra}(\hat{T},T) = \sum_{x,y} \left\| |\hat{T}_{x,y} - \hat{T}_{x-1,y}| - |T_{x,y} - T_{x-1,y}| \right\|_1 + \sum_{x,y} \left\| |\hat{T}_{x,y} - \hat{T}_{x,y-1}| - |T_{x,y-1} - T_{x-1,y-1}| \right\|_1 \tag{5}$$

Here x and y denote the pixel value of the video frame.

3.4 Objective Function

The weighted sum of the adversarial constraints and the constraints on motion and structural similarity constitutes the objective function. The objective function needs to be minimized as shown below. The objective function for the generator training step:

$$L_{\mathcal{G}} = \lambda_{\text{int}}\, L_{int}\left(\hat{T}_{r+1}, T_{r+1}\right) + \lambda_{gra} L_{gra}\left(\hat{T}_{r+1}, T_{r+1}\right) + \lambda_{adv} L_{adv}^{\mathcal{G}}\left(\hat{T}_{r+1}\right) \tag{6}$$

The objective function for the discriminator training step:

$$L_{\mathcal{D}} = L_{adv}^{\mathcal{D}}\left(\hat{T}_{r+1}, T_{r+1}\right) \tag{7}$$

The parameters λ_{int}, λ_{gra}, and λ_{adv} are the coefficients that maintain the weights symmetry among the loss functions.

3.5 Anomaly Score

Peak-Signal-to-Noise Ratio (PSNR), which acts as a metric evaluating the quality of a video frame is used to calculate the anomaly score. It measures the disparity between the real and reconstructed video frames, with higher PSNR values signifying lower distortion.

$$\text{PSNR}(T,\hat{T}) = 10\log_{10} \frac{\left[\max_{\hat{T}}\right]^2}{\frac{1}{N}\sum_{j=0}^{N}\left(T_j - \hat{T}_j\right)^2} \tag{8}$$

Here, T and $\hat{T}$ represent the real video frame and the reconstructed video frame. A high PSNR value for the future video frame suggests a higher likelihood of it being considered normal. PSNR for each frame is computed in every testing video subsequently, we normalize the PSNR values of all video frames within each testing video (AS) to a range between 0 and 1. The anomaly score for the predicted video frame is then determined using the following equation:

$$AS = \frac{PSNR\left(T_k, \hat{T}_k\right) - \min_k \text{PSNR}\left(T_k, \hat{T}_k\right)}{\max_k \text{PSNR}\left(T_k, \hat{T}_k\right) - \min_k \text{PSNR}\left(T_k, \hat{T}_k\right)} \tag{9}$$

Hence, the prediction of a frame being normal or abnormal relies on its assigned score, denoted as AS. By establishing a threshold, one can effectively differentiate between normal and anomaly video frames.

4 Experimental Evaluation and Results

In this section, we compare the proposed iMAppGAN model to other state-of-the-art(SOTA) AD methods. The following benchmark video AD datasets are taken: CUHK Avenue [16], Shanghaitech [18], and UCSD Ped2 [20]. The test videos in each dataset contain both normal and anomalous frames, while the training videos are comprised of normal frames only. Frame level annotations serve as flag bits in the test frames, where a 1-bit designates an anomalous frame, and a 0-bit denotes a normal video frame. The video frame considered for experimentation consists of 10 video frames each of size 160×160 converted into grayscale. We train iMAppGAN using the Adam optimizer with a momentum of 0.9 a decay rate of 10^{-4} and a batch size of 128. The number of epochs for UCSD Peds2, CUHK Avenue, and ShanghaiTech are 80, 80, and 90 respectively. All tests are performed on a dedicated GPU server that runs Ubuntu Mate 16.04 and has an Intel Xenon Gold 6226R processor clocked at 2.9 GHz, 128 GB of RAM, and a V100 Nvidia TESLA GPU. We utilize the TensorFlow library (version 2.13) to implement our video AD architecture.

4.1 Datasets

This section focuses on the benchmark video AD datasets to evaluate our proposed method. The video frame instances for both the normal and anomaly samples from the benchmark video AD datasets are shown in Fig. 4.

UCSD Peds2 [2]**:** It is based on a fixed camera position at a height and looking down on pedestrian routes. Bikers, skateboarders, small carts, and pedestrians crossing a path or in its vicinity are common anomalies. The UCSD Ped2 dataset comprises 12 test video clips and 16 training video clips.

CUHK Avenue [16]**:** It includes RGB videos, encompassing 16 training videos with a collective video frame count of 15,328. Additionally, there are 21 testing videos, totaling 15,324 frames. The size of each video frame is 360×640. The dataset contains 47 abnormal events, such as running, loitering, and walking in opposing directions.

Shanghaitech Dataset [18]**:** The Shanghaitech dataset consists of an extensive crowd-counting dataset, incorporating 13 scenes characterized by diverse light conditions and camera perspectives. For training purposes, the dataset offers a considerable resource, comprising over 270,000 frames encompassing 130 distinct abnormal events. Notably, it is augmented by pixel-level ground truth annotations for abnormal events, significantly enhancing its value for research and analysis.

4.2 Pre-processing and Anomaly Detection Strategy

The input video frame is resized to 160×160 pixels and the OpenCV library is utilized to extract video frames from the videos. Further, the normalization of pixel intensity within the $[-1, 1]$ range is conducted before training and evaluating the model. The normalized PSNR score is utilized to differentiate between

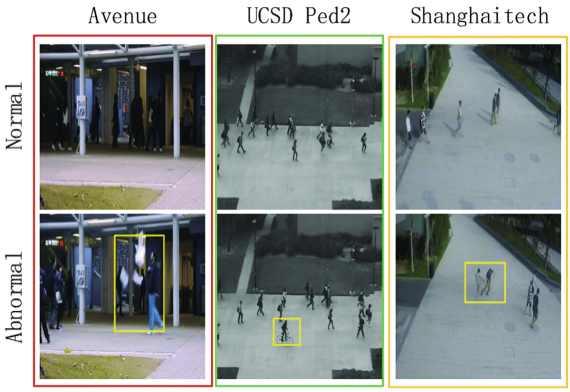

Fig. 4. The normal and abnormal video frames in the first column are from the CUHK Avenue dataset, whereas the second column is from the UCSD Ped2 dataset, and the third column is from the ShanghaiTech dataset.

the normal video frames and anomalous video frames using the thresholding method. The resulting score is compared against the threshold values to decide if the video frame is normal or anomaly. The threshold values for UCSD Ped2, CUHK Avenue, and ShanghaiTech datasets are 0.71, 0.62, and 0.86 respectively.

4.3 Evaluation Metrics

We used the Area under the curve (AUC) of the Receiver Operating Characteristics (ROC) score and the Equal error rate (EER) to evaluate our model. The ROC curve provides insights into the model's performance in distinguishing between normal and anomalous instances to evaluate the proposed model on video AD.

Further, we have analyzed the benchmark datasets on our model iMAppGAN at first by considering only the reconstruction network and then only the prediction network. The results of all these tests are shown in Table 2, and it can be seen that the performance of iMAppGAN is superior to the only reconstruction or only prediction network.

1. **AUC Score**
   The AUC score is a metric used to evaluate the performance of a classification model, which depicts the trade-off between a True Positive Rate (TPR) and a False Positive Rate (FPR). A high AUC score which is closer to 1, signifies better model discrimination and performance.
2. **EER score**
   In video AD, the EER metric is employed to assess the model's capability to balance false positives and false negatives. It signifies the point on the ROC curve where the false acceptance rate equals the false rejection rate. A lower EER in video anomaly detection implies a more optimal trade-off,

Table 1. Comparison of the EER and AUC score for Avenue, UCSD Ped2 and ShanghaiTech Dataset

	UCSD Ped2		CUHK Avenue		Shanghaitech	
Models	AUC	EER	AUC	EER	AUC	EER
FFP + MC [15]	95.4	–	85.1	–	72.8	–
sRNN-AE [19]	92.2	–	83.5	–	69.6	–
3DAE + ADL [35]	94.9	10.3	86.1	22.0	–	–
STU-Net [14]	96.5	8.7	84.5	21.5	–	–
SCLSTM AE [30]	89.5	14.2	79.7	23.0	67.2	31.5
ISTL [22]	91.1	8.9	76.8	29.2	–	–
MemAE [7]	94.1	–	83.3	–	71.2	–
Ada-Net [28]	90.7	11.5	89.2	17.2	70.0	36.5
CCGAN [23]	95.5	11.0	–	–	–	–
IPR [29]	96.3	10.0	85.1	–	73.0	–
FFP+MS_SSIM+FCN [33]	95.9	11.1	85.9	20.4	73.9	–
Clustering AE [3]	96.5	–	86.0	–	73.3	–
MESDnet [6]	95.6	–	86.3	–	73.2	–
STemGAN [25]	97.5	6.3	86.0	20.1	–	–
MONAD [5]	97.2	–	86.4	–	70.9	–
VALDGAN [26]	97.7	7.01	91.03	9.04	–	–
ASTNet [12]	97.4	10.2	86.7	20.3	73.6	–
A2D-GAN [27]	97.4	5.1	91.0	9.0	74.2	25.2
iMAppGAN(Ours)	97.9	5.9	90.8	10.7	75.3	25.4

Table 2. Comparative analysis of Reconstruction model, Prediction model, and iMAppGAN

Datasets	iMAppGAN	Reconstruction	Prediction
UCSD ped2	97.9	95.5	96.8
CUHK Avenue	90.8	84.6	86.7

indicating improved overall performance in distinguishing between normal and anomalous events.

4.4 Comparison Study

The performance of our model has been evaluated through a comparison with other SOTA techniques on three benchmark video AD datasets. Our proposed model outperforms other SOTA models. Table 1 shows the comparison results of the proposed method with other SOTA models based on the AUC score. As can be drawn from Table 1 our proposed method obtains AUC and EER scores

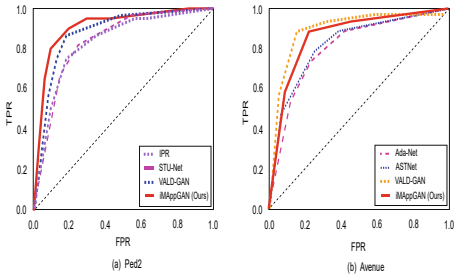

Fig. 5. ROC curve comparison of various SOTA methods on CUHK Avenue dataset and UCSD Ped2 dataset.

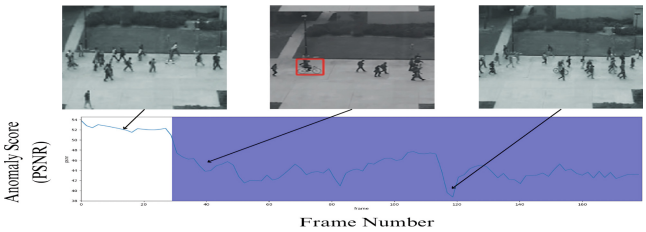

Fig. 6. PSNR Score for a video in UCSD ped2. The purple region in the graph shows an anomaly. In this figure when the biker enters the frame the anomaly score starts decreasing and attains minima, thereby depicting anomaly.

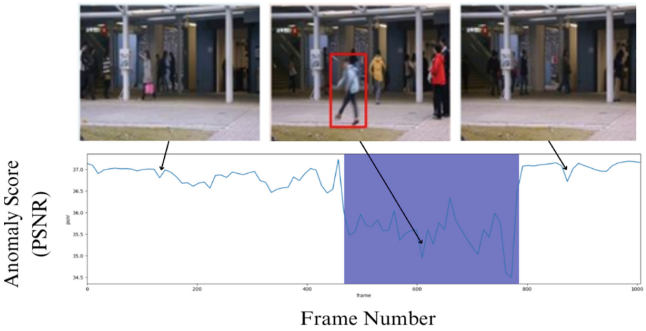

Fig. 7. Visualization of Anomaly Score for a video in CUHK Avenue. The purple region in the graph shows where an anomaly. In this figure when a running person enters the frame the anomaly score starts decreasing and attains minima, thereby depicting the anomaly and it goes back to normal after the anomaly vanishes from the video.

of 97.9% and 5.9% on the UCSD ped2 dataset, 90.8% and 10.7% on the CUHK avenue dataset, and 75.3% and 25.4% on ShanghaiTech dataset respectively.

We examine the obtained results below. Moreover, using the TPR and FPR readings at various thresholds, the receiver operating characteristic (ROC) curve for UCSDped2 on our model as displayed in the Fig. 5 is plotted by us. Figure 6

and 7 show the plot of PSNR scores of a video from USCD ped2 and CUHK avenue datasets, respectively.

5 Conclusion and Future Scope

In this work, we proposed a novel Integrated Motion-Appearance Generative Adversarial Network, which utilizes the temporal shift technique to capture both motion and appearance features. The model predicts future video frames from a sequence of input video frames using channel attention, which is then reconstructed. Additional structural-similarity constraints are integrated during the reconstruction process to enhance the generation of normal video frames. Moreover, the discriminator network is employed to facilitate the training of the generator network. To assess whether a video frame is normal or anomalous, we use the Peak Signal-to-Noise Ratio (PSNR) by comparing the generated video frame with the real video frame.

Our model outperforms other state-of-the-art models in terms of AUC (Area Under Curve) and EER (Equal Error Rate) on benchmark video anomaly detection datasets. The methodology can be extended to handle and generate video cubes instead of single video frames. Furthermore, we address the absence of models specifically designed for systematically classifying anomalies into distinct categories by proposing a classification model dedicated to identifying and categorizing anomalies based on their unique features.

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